Effect of Mobility on Network Performance in Wi-Fi and MANET

Nikitha Baddam
University of Nebraska–Lincoln
nbaddam2@huskers.unl.edu

Under the guidance of
Professor Byrav Ramamurthy
School of Computing, University of Nebraska–Lincoln
ramamurthy@unl.edu

Abstract

Mobile Ad-Hoc Networks (MANETs) allow mobile devices to communicate without depending on existing infrastructure such as access points or routers. Their flexibility makes them highly useful in environments where traditional networks cannot be deployed. However, node mobility introduces frequent route breakages, which impact performance. In this project, we simulate a 20-node MANET using ns-3 with the AODV routing protocol and Random Waypoint mobility model. We vary mobility speeds between 1–25 m/s and evaluate throughput, packet delivery ratio (PDR), and end-to-end delay. Our observations show that performance gradually decreases as mobility increases due to unstable routes and repeated rediscovery processes.

1 Introduction

1.1 Mobile Ad-Hoc Networks Overview

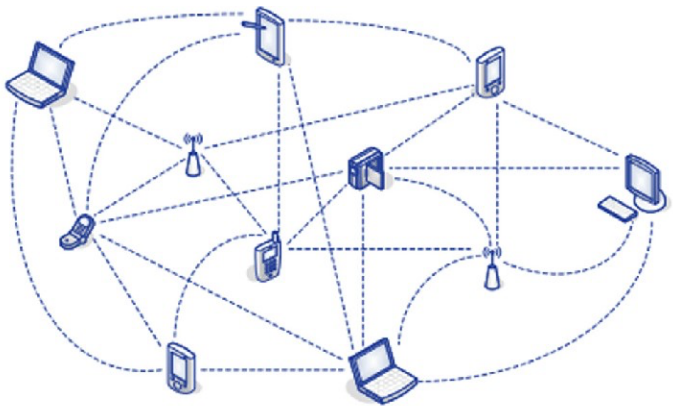

Figure 1: Basic structure of a MANET.

Mobile Ad-Hoc Networks (MANETs) are decentralized, self-organizing networks where mobile nodes communicate wirelessly. Unlike Wi-Fi networks, which depend on fixed infrastructure, MANETs rely on node-to-node cooperation. Each device forwards data on behalf of others, making the network adapt quickly as nodes move. Because the topology keeps changing, maintaining reliable communication becomes challenging.

MANETs are widely used in:

- Disaster and emergency response operations
- Military and defense communication systems
- Vehicular and transportation networks (VANETs)
- Remote sensor and environmental monitoring

1.2 The AODV Routing Protocol

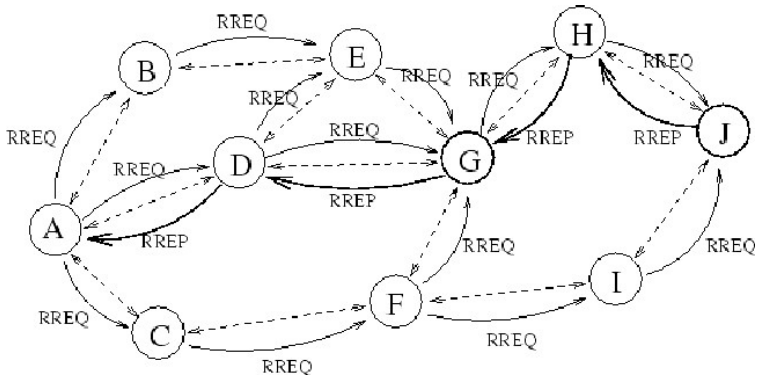

Figure 2: AODV route discovery process.

2 AODV (Ad-hoc On-Demand Distance Vector) is a reactive routing protocol. Instead of maintaining routes all the time, it creates them only when needed. When a source node wants to send data, it broadcasts a Route Request (RREQ). Nodes forward this request until the destination or an intermediate node replies with a Route Reply (RREP). If a link breaks, AODV sends a Route Error (RERR) and starts a new discovery.

Additional insights:

- AODV reduces overhead by maintaining routes only when required.
- It works well in networks with low to moderate mobility.

1.3 Project Objective

The main objective of this project is to understand how node mobility affects MANET performance using ns-3 simulations. Specifically, we aim to:

- Evaluate how throughput, PDR, and delay change as node speed increases from 1–25 m/s.
- Examine how often AODV routes break and how quickly they recover.
- Understand the impact of mobility on the stability of multi-hop paths.

2 Simulation Setup

2.1 Network Parameters

Table 1: Simulation Parameters

Parameter	Value
Number of nodes	20
Network area	1000 × 1000 m
Routing protocol	AODV
Mobility model	Random Waypoint
Speeds tested	1–25 m/s
Traffic type	UDP CBR
Packet size	512 bytes
Simulation time	150 seconds

2.2 Mobility Model

The Random Waypoint (RWP) model is used to simulate node movement. In this model, each node:

- Selects a random destination point within the simulation area.
- Moves toward that point at a speed between 1–25 m/s.
- Pauses briefly before choosing the next destination.
- Continues to repeat this pattern throughout the simulation.

2.3 Traffic and Performance Metrics

CBR traffic is generated from Node 0 to Node 19. FlowMonitor is used to calculate:

- **Throughput** – amount of data successfully delivered per second.
- **Packet Delivery Ratio (PDR)** – percentage of packets that reach the destination.
- **Delay** – time taken for packets to travel end-to-end.

- Routing stability, based on how frequently AODV rebuilds routes.
- Impact of increasing mobility on packet loss and retransmissions.

3 Results

- The table summarizes how throughput, PDR, and delay vary for node speeds between 1–25 m/s.
- It provides a clear comparison showing how performance decreases as mobility increases.

Speed (m/s)	Throughput (kbps)	PDR (%)	Avg Delay (ms)
1	513.33	98.66	1.28
2	513.30	98.65	1.04
3	513.33	98.66	0.97
4	505.07	97.07	1.06
5	351.65	67.59	0.99
6	295.37	56.77	1.02
7	505.67	97.19	1.45
8	218.02	41.90	1.10
9	510.42	98.10	1.57
10	507.80	97.60	2.65
11	505.53	97.16	1.34
12	271.12	52.11	0.98
13	500.49	96.19	5.67
14	497.55	95.63	5.87
15	300.93	57.84	3.94
16	483.32	92.89	4.99
17	344.97	66.30	9.76
18	176.14	33.85	6.41
19	333.25	64.05	8.83
20	376.42	72.35	5.69
21	220.55	42.39	1.80
22	234.49	45.07	10.49
23	420.54	80.83	8.82
24	350.53	67.37	8.48
25	341.34	65.60	6.83

Figure 3: Performance results for speeds 1–25 m/s.

3.1 Throughput vs Speed

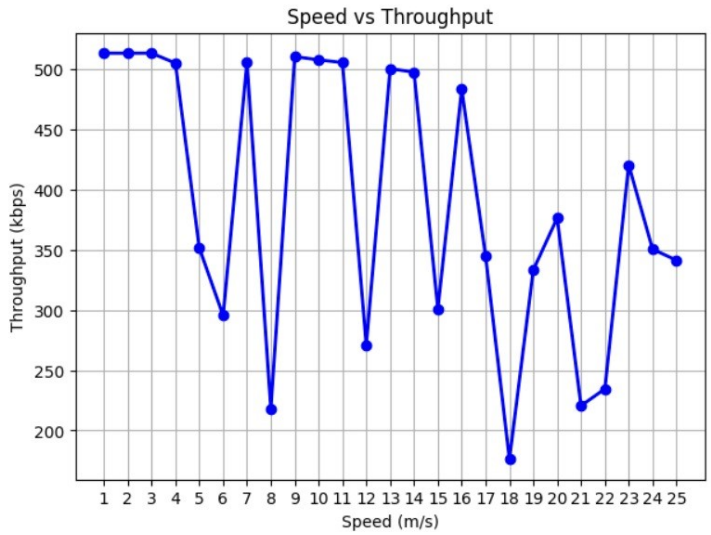

Figure 4: Throughput vs node speed

As node speed increases, throughput gradually decreases. At low speeds, routes remain stable, so packets flow smoothly. But at higher speeds (above 10–15 m/s), links break often, forcing AODV to recreate routes repeatedly. This disrupts data flow and lowers the total number of packets successfully delivered per second.

3.2 PDR vs Speed

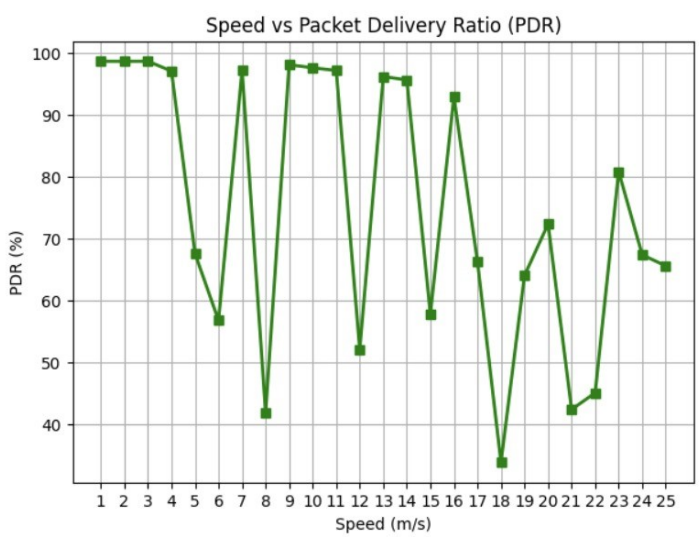

Figure 5: PDR vs node speed

Packet Delivery Ratio (PDR) drops as mobility increases. When nodes move faster, packets are more likely to be lost during transmission because the route formed by AODV quickly becomes outdated. This leads to more dropped packets before they reach the destination.

3.3 Delay vs Speed

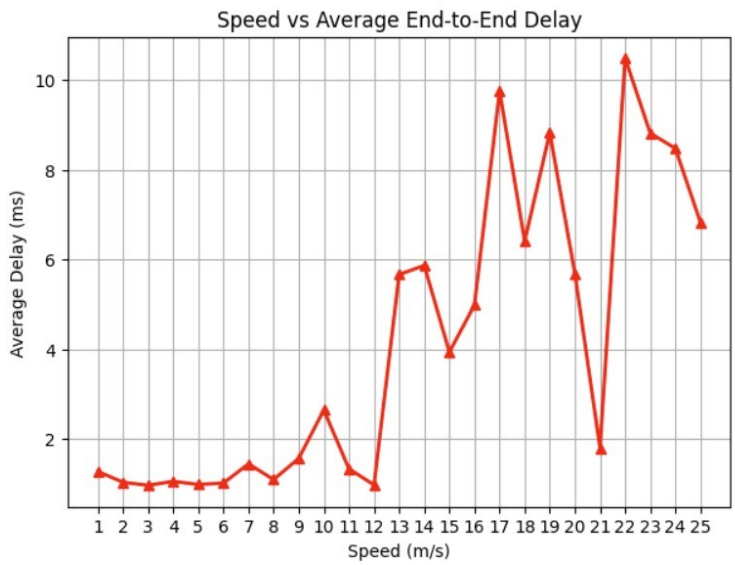

Figure 6: End-to-end delay vs node speed

End-to-end delay increases sharply at high speeds. This is because AODV must frequently initiate route discovery whenever links break. Each new route search adds extra time, causing packets to wait longer before being delivered.

4 Discussion

AODV performs efficiently at low speeds, but as mobility increases, its route maintenance process struggles to keep up. High-speed movement produces rapid topology changes, causing packets to be dropped and routes to be recalculated frequently. This explains the observed decline in performance across all metrics.

5 Conclusion

Based on the simulation, we conclude:

- Mobility has a direct and significant effect on MANET performance.
- Throughput and PDR drop steadily as node speed increases.

- Delay increases because AODV must frequently rebuild routes.
- AODV is better suited for low-mobility environments than high-mobility ones.
- For highly dynamic scenarios, more adaptive routing protocols are necessary.

6 Future Work

- Compare AODV with DSR, OLSR, and hybrid protocols such as ZRP.
- Use alternative mobility models such as Gauss–Markov and Manhattan Grid.
- Analyze larger network sizes (50–200 nodes) for scalability.
- Introduce TCP traffic to compare performance with UDP flows.

Acknowledgment

We would like to thank Professor Byrav Ramamurthy for his guidance, support, and feedback throughout the project. His insights helped us understand MANET behavior more clearly and complete this work successfully.

References

- ns-3 Network Simulator Documentation, 2024.
- Perkins, C. “AODV Routing Protocol,” IETF RFC 3561.
- S. Corson and J. Macker, “Mobile Ad hoc Networking (MANET): Routing Protocol Performance Issues.”